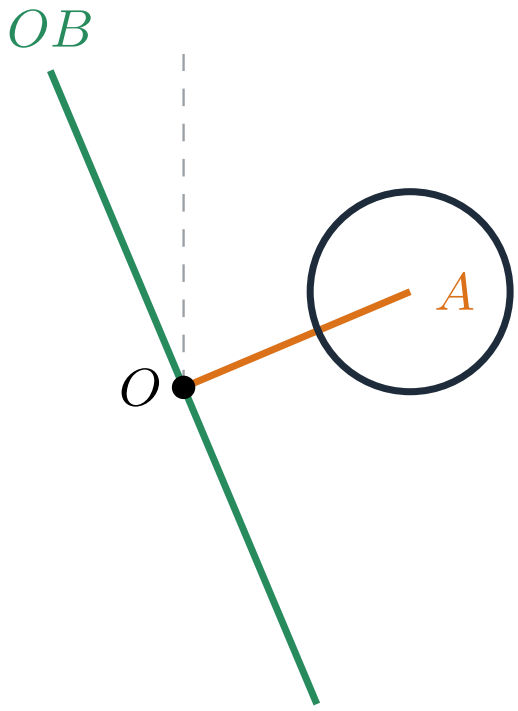
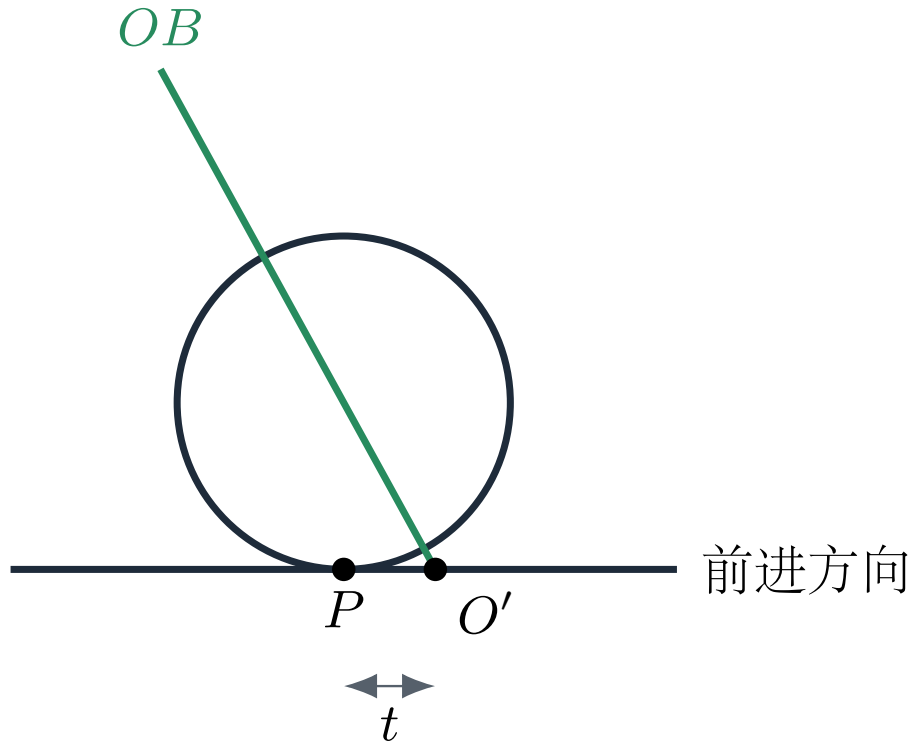


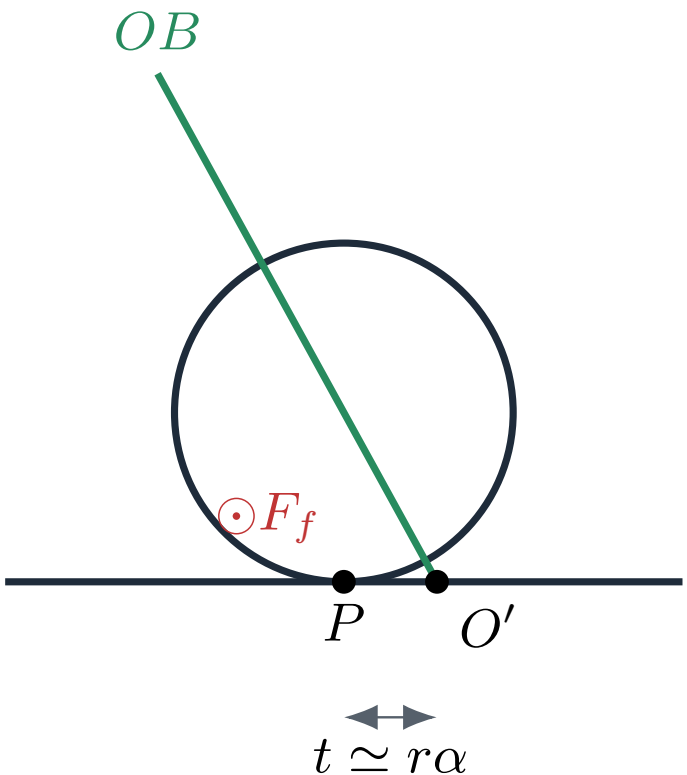
正中位置：$OA \perp OB$

OA 随主销内倾而倾斜



后倾的侧视图

轴与地面的交点 O' 在 P 前方

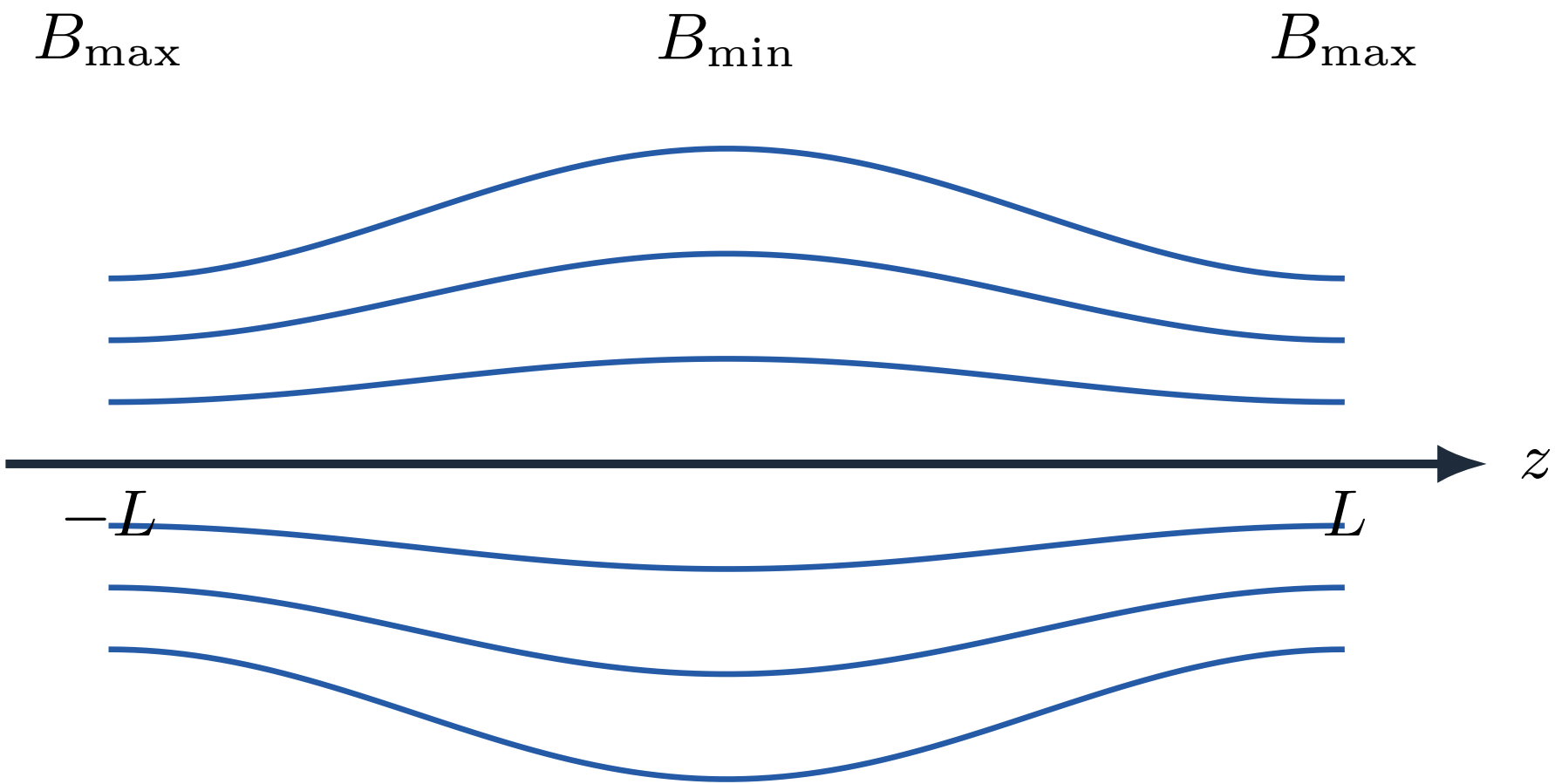


侧向力垂直纸面，不能画成沿地面的力

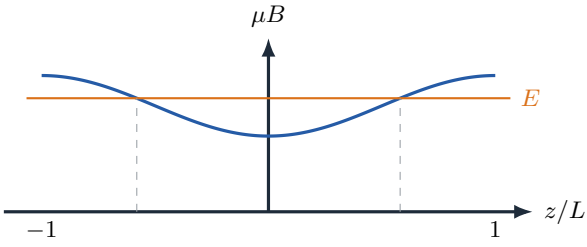
主销内倾：$M_1 = -Gl \sin\beta \sin\theta/4$

后倾：$F_f \simeq mv^2\theta/(2d)$

$M_2 \simeq -F_f r\alpha$



中心束较宽、磁场较弱；两端束较窄、磁场较强

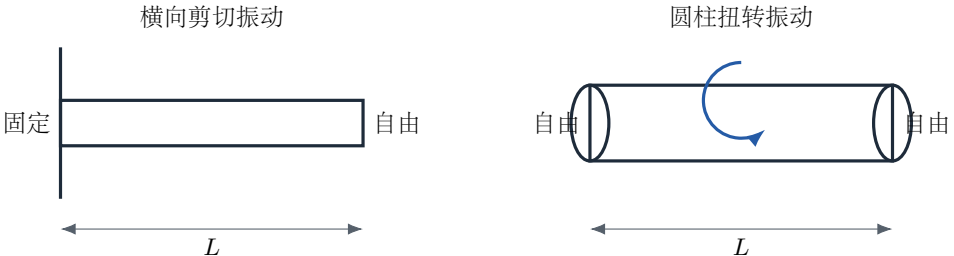


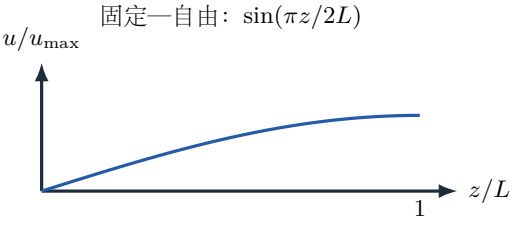
$K_{\parallel} = E - \mu B(z)$

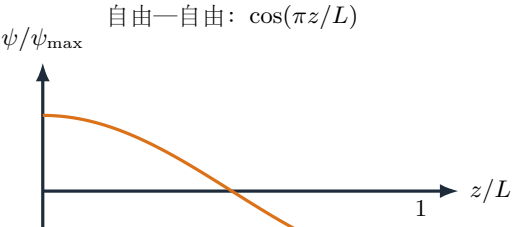
$E < \mu B_{\max}$：两端之前反射

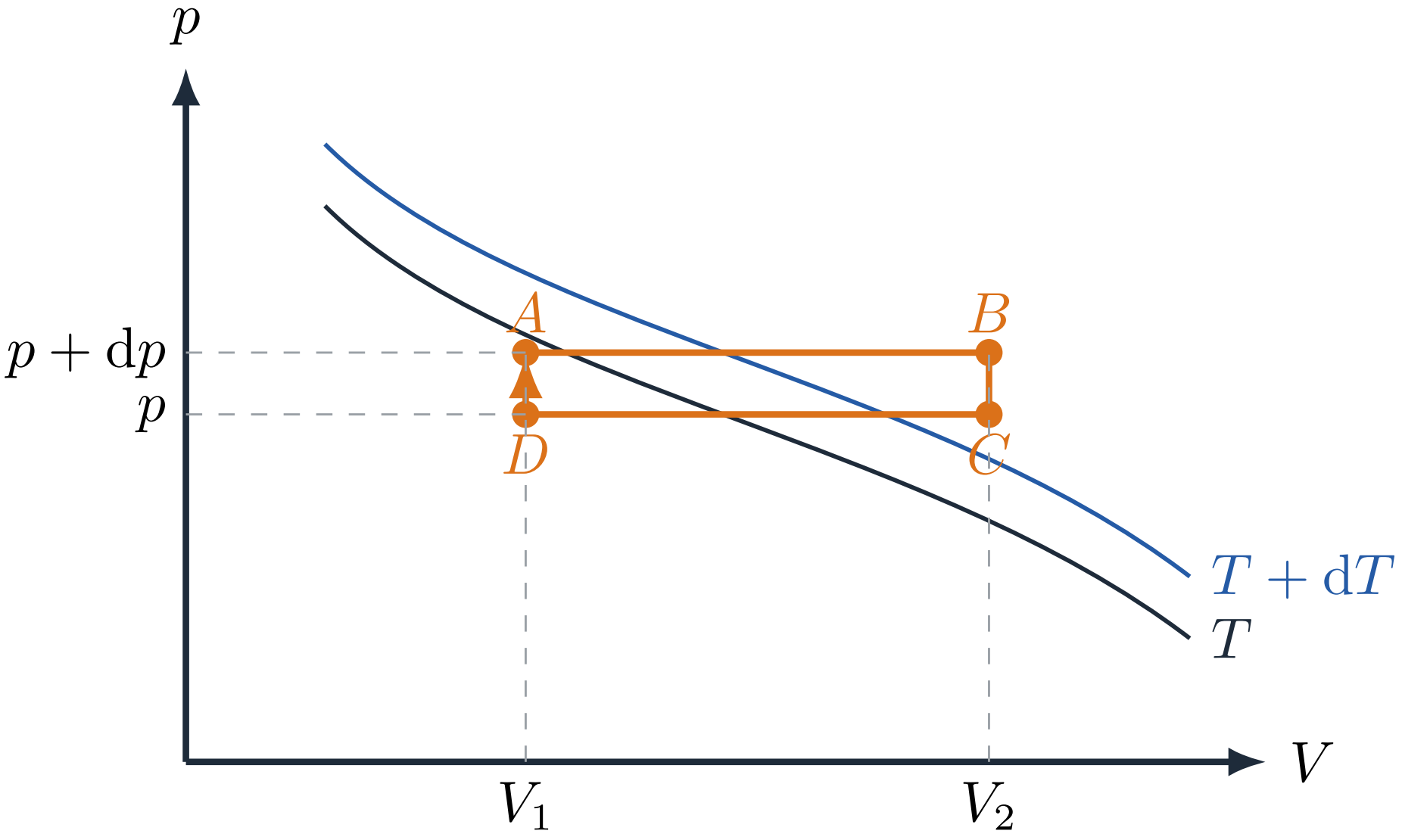
$E > \mu B_{\max}$：有限时间到达端点

光滑最大值处的等号可为渐近极限


横向剪切振动
固定
自由
L
圆柱扭转振动
自由
自由
L


固定—自由：$\sin(\pi z/2L)$
$u/u_{\max}$
z/L
1


自由—自由：$\cos(\pi z/L)$
$\psi/\psi_{\max}$
z/L
1


p
$p + \mathrm{d}p$
p
A
B
D
C
$T + \mathrm{d}T$
T
V
V_1
V_2

逐字采用题设 $M=-\mu_0 B$：

$\Delta S=-\mu_0 B_c B_c'$

形式数值约 4.08×10^{-9}

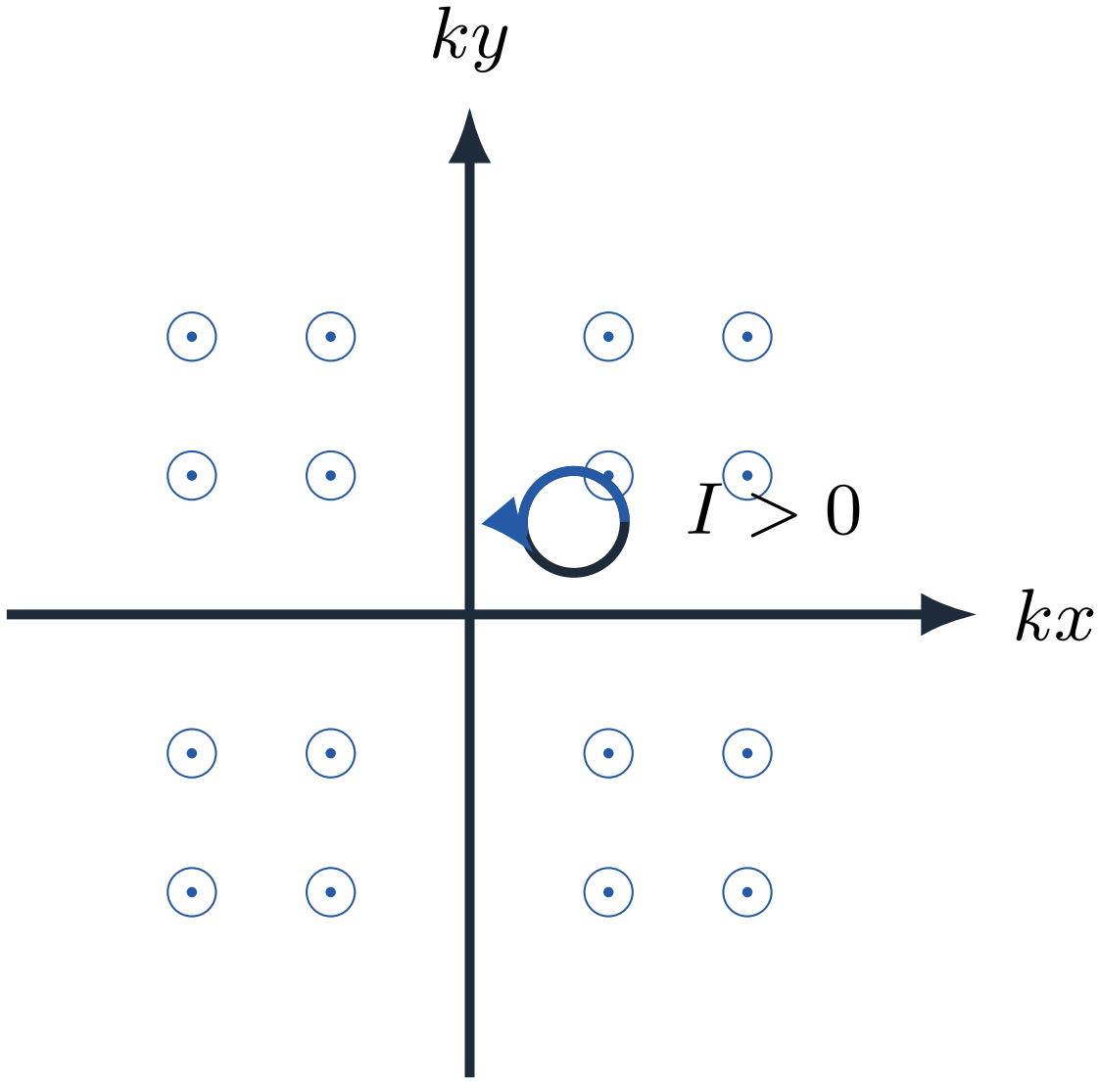
与题给 SI 磁导率单位不相容；不得标成 J/K

改用 SI 磁化强度 $M=-B/\mu_0$：

$\Delta s=-B_c B_c'/\mu_0$

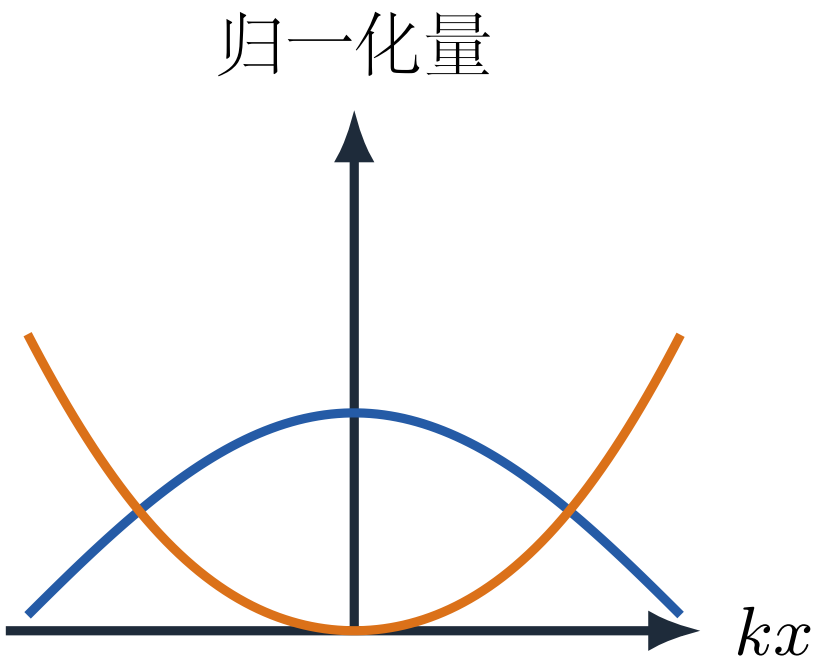
$\Delta s\simeq 2.61\times10^{3}\ \mathrm{J\,m^{-3}\,K^{-1}}$

总熵变还需要样品体积，原题未给



$B = B_0 \cos kx \cos ky$；本图限于 $|kx|, |ky| < \pi/2$

外部区域还会出现反向磁场，小环电流取逆时针为正



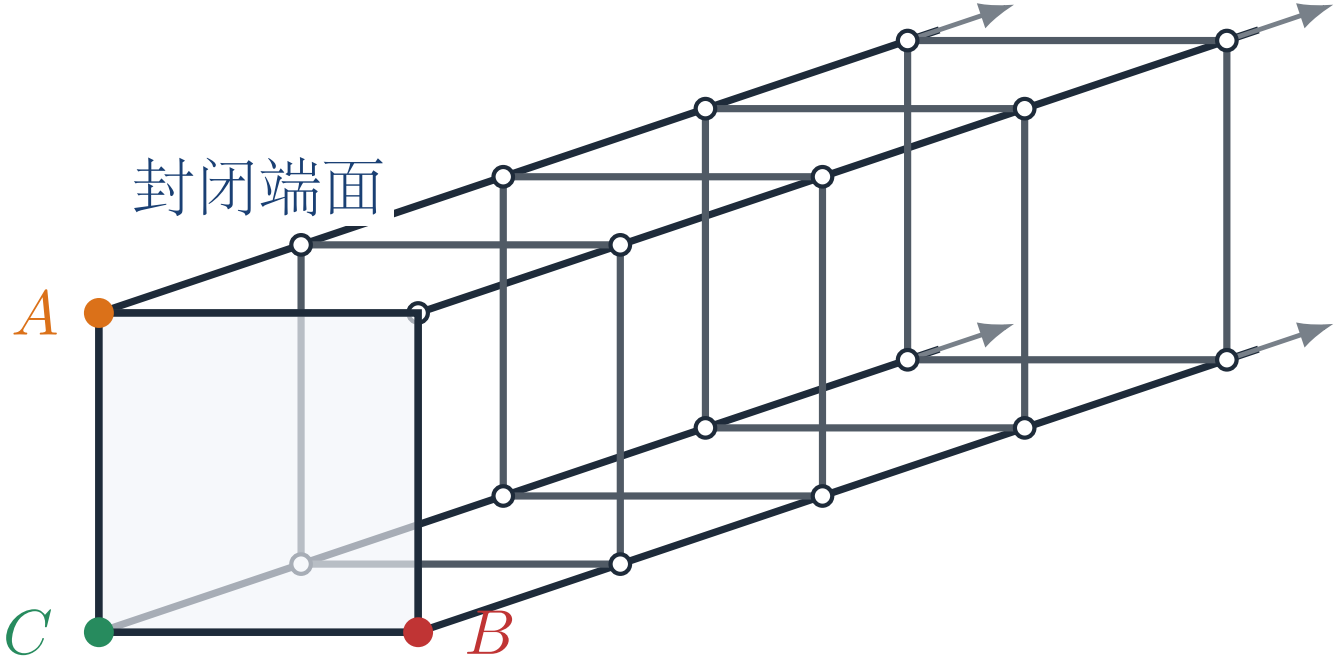
取 $y=0,\ LI_0=B_0S$ 作图

蓝：$B/B_0=\cos kx$

橙：$(U-U_0)/(I_0B_0S)$

$U=(\Lambda-SB)^2/(2L)$

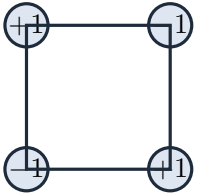
原点是小振动平衡点



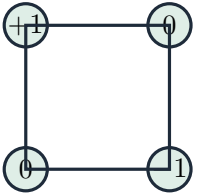
向右后方无限延伸

$n \to \infty$

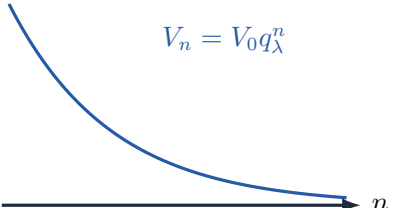
所有横向边与纵向边的电阻均为 r



$\lambda = 4$



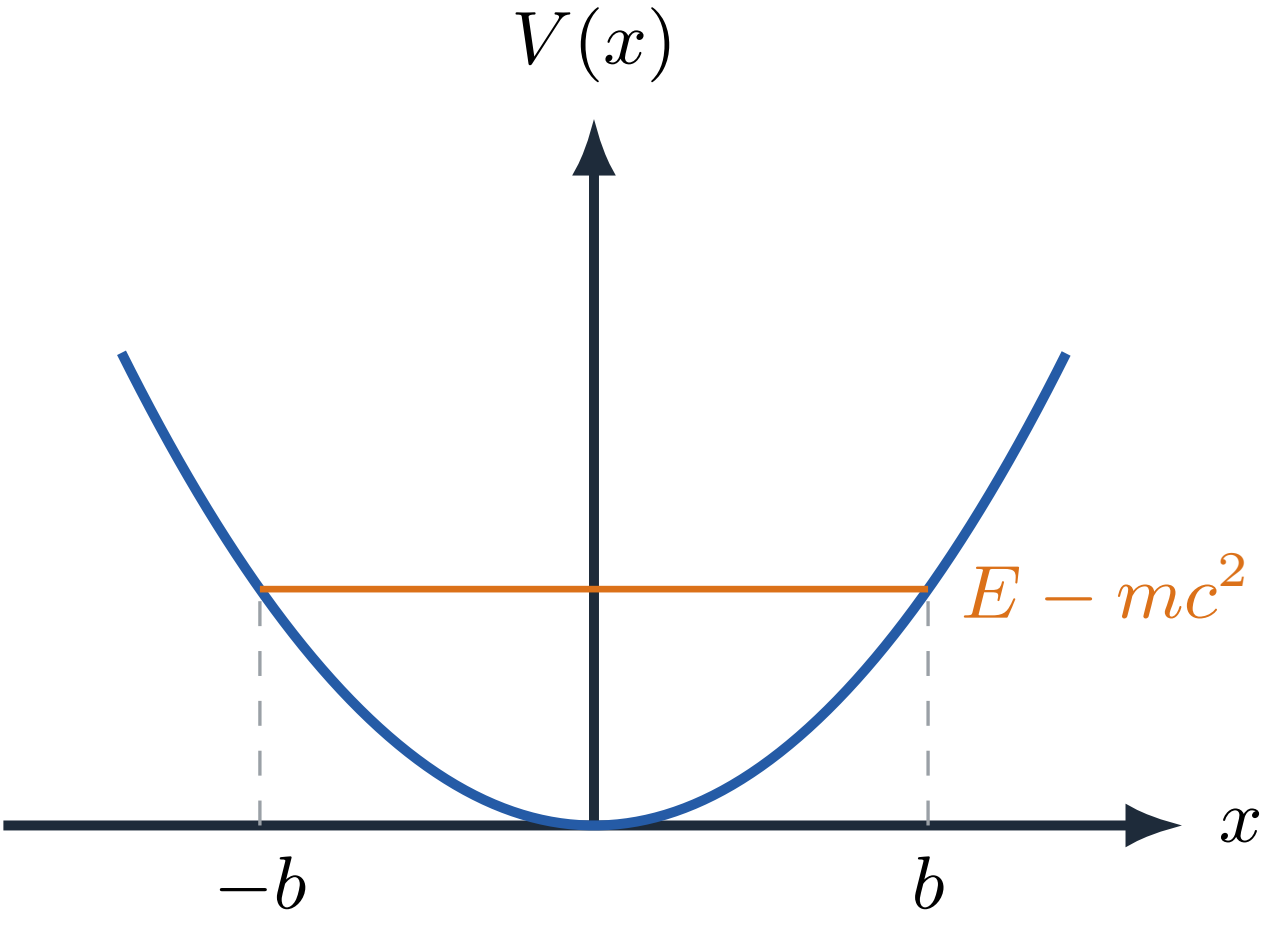
$\lambda = 2$

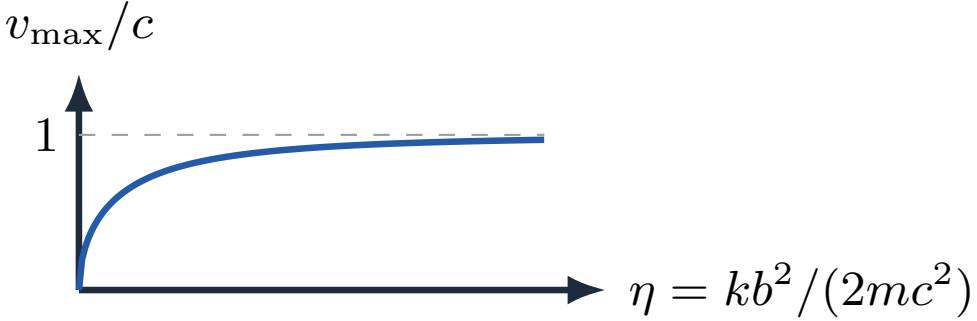


$q_2 = 2 - \sqrt{3}$

$q_4 = 3 - 2\sqrt{2}$

转折点满足 $V(b) = E - mc^2$

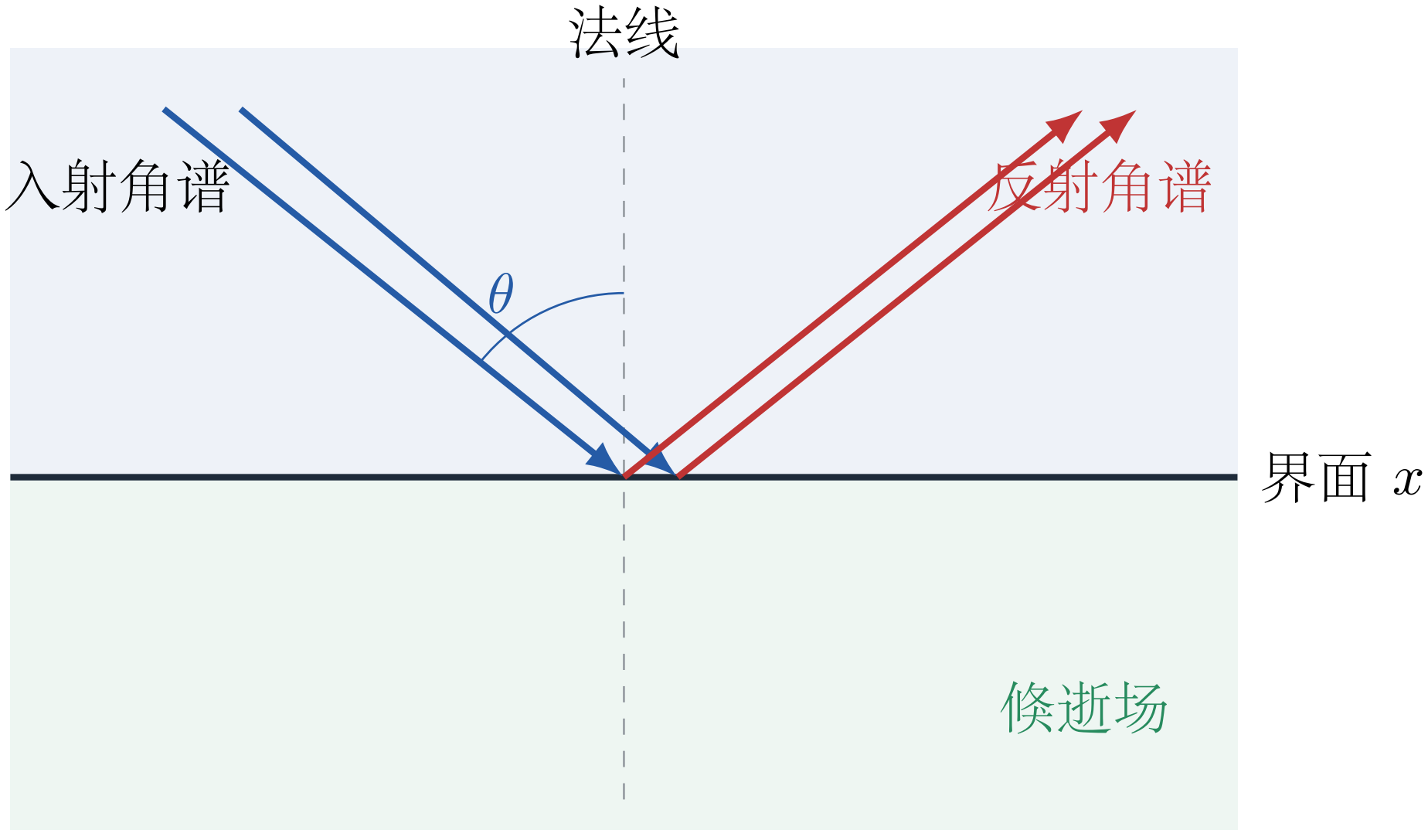


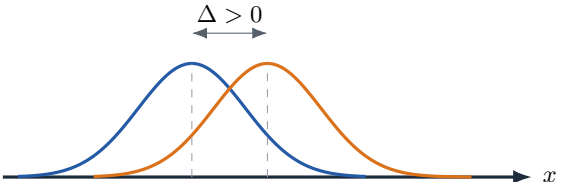


$$v_{\max}/c = \sqrt{1-(1+\eta)^{-2}}$$

$$T = 2\pi\sqrt{m/k}\,[1+3\eta/8+O(\eta^2)]$$

一阶周期公式只用于 $\eta \ll 1$


法线
入射角谱
反射角谱
θ
界面 x
倏逝场



$r = e^{i\delta}$，场约定 $e^{i(\omega t - \boldsymbol{k}\cdot\boldsymbol{r})}$

反射实场：$\cos(k_x x - \omega t - \delta)$

$\Delta = \delta'(\theta)/(k\cos\theta)$

蓝：几何反射峰；橙：带相位的波包峰